

Time-Domain Adversarial Attacks on I/Q-Based LPI Radar Detection

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ABSTRACT

Research on Low Probability of Intercept (LPI) radar detection and recognition has predominantly followed a pipeline that converts signals into time-frequency images (TFIs) and then applies deep learning for classification. More recently, detectors that directly ingest time-domain in-phase/quadrature (I/Q) signals without image conversion have been proposed. Aligning with this trend, we systematically analyze the vulnerability of I/Q-based LPI detection and modulation recognition to time-domain adversarial attacks under perturbation-to-signal ratio (PSR) and perturbation-to-noise ratio (PNR) constraints. Specifically, we apply the Fast Gradient Sign Method (FGSM), Projected Gradient Descent (PGD), and their momentum- and diversity-enhanced variants (MI-FGSM and MI-DI-FGSM) to twelve representative LPI modulation classes, and evaluate per-modulation probability of detection and attack success rate (ASR) to characterize sensitivity. The results establish a robustness baseline for time-domain I/Q LPI intelligent receivers and provide practical guidance on attack and defense configurations across feasible operational conditions. We further observe clear modulation-dependent behavior, with polyphase and T-code families exhibiting higher vulnerability and predominantly within-family confusions, while Costas and BPSK remain comparatively robust.

Keywords : LPI radar, Adversarial examples, In-phase/Quadrature (I/Q), FGSM, PGD

I. Introduction

Low Probability of Intercept (LPI) radars are engineered to conduct reconnaissance, target acquisition, and situational awareness while remaining outside an adversary's detection and monitoring networks [1]. As such, they are regarded as a critical capability in modern military operations. The adoption of LPI techniques is rapidly expanding across electronic warfare (EW), intelligence-surveillance-reconnaissance (ISR), stealth platforms,

unmanned systems, and maritime/ground surveillance [1]. In these domains, the ability to evade interception has become a decisive factor for survivability and mission success. This wider and more sophisticated operational use of LPI radar has, in turn, accelerated technical efforts to detect, identify, and classify such emissions.

In response to these trends, deep-learning-based LPI detection has advanced rapidly. A dominant paradigm converts the received in-phase/quadrature (I/Q) into a time-frequency

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image (TFI) and processes it with convolutional neural networks (CNNs) [2–4]. By mapping diverse LPI waveforms into a common two-dimensional representation, TFIs reveal salient intra-pulse signatures that are convenient for feature learning, enabling joint detection and modulation recognition with high accuracy and throughput. Consequently, much of the prior literature has centered on the TFI-based deep learning pipeline [5].

In parallel, time-domain I/Q intelligent receivers (models that take raw complex baseband directly) are being deployed in a range of operational settings [6], [7]. Direct I/Q ingestion reduces dependence on front-end preprocessing choices (e.g., short-time Fourier transform (STFT) parameters, normalization, or dB clipping), which can significantly affect the resulting time-frequency representation and consequently influence classification performance. For example, parameters such as the STFT window length, overlap ratio, or kernel settings in alternative time-frequency distributions (e.g., Wigner–Ville distribution (WVD) or Choi–Williams distribution (CWD)) can alter the spectral structure of the representation, making it difficult to design models that generalize robustly across different preprocessing configurations and signal processing pipelines [8].

In contrast, direct I/Q ingestion allows models to operate directly in the signal domain and facilitates the incorporation of physical constraints—such as transmit/receive power budgets, error vector magnitude (EVM), and out-of-band emissions—within the input space. Although TFI-based detectors remain the majority in the LPI literature, I/Q-based detection provides a practical alternative, as TFI approaches require a mandatory signal-to-image conversion stage that can

impose nontrivial constraints on real-time operation.

Under these circumstances, a systematic assessment of adversarial vulnerability for I/Q-based LPI detectors is essential. While several studies have shown that small input perturbations can reduce probability of detection or induce misclassification in TFI-based systems [9], comparable analyses for I/Q-domain LPI detection are largely absent. In particular, there remains limited evidence regarding (i) the extent of evasion achievable by time-domain perturbations that comply with realistic perturbation-to-signal ratio (PSR) and perturbation-to-noise ratio (PNR) constraints [10], and (ii) whether consistent evasion can be maintained across attack methods and modulation types.

Motivated by these considerations, this paper presents a systematic study of adversarial attacks on I/Q-based LPI signal classification. We investigate the effectiveness of multiple gradient-based attacks, including Fast Gradient Sign Method (FGSM) [11], Projected Gradient Descent (PGD) [12], Momentum Iterative FGSM (MI-FGSM) [13], and Momentum Diverse Input FGSM (MI-DI-FGSM) [14], directly in the raw signal domain. Furthermore, we extend input diversity techniques—originally developed for image-based attacks—to the signal domain by designing transformations tailored to I/Q sequences.

In addition to evaluating attack strength under various PSR/PNR constraints, we analyze adversarial robustness across different model architectures and representation domains. Specifically, we compare raw signal-based detectors with TFI-based models and examine both white-box and black-box attack scenarios.

The main contributions of this paper are summarized as follows:

- Comprehensive adversarial evaluation: We provide the first systematic evaluation of adversarial vulnerability in I/Q-based LPI signal classification across a wide range of SNR and PSR/PNR conditions.
- Advanced attack strategies: We extend conventional FGSM and PGD attacks by incorporating momentum and input diversity (MI-FGSM and MI-DI-FGSM) in the signal domain, demonstrating significantly improved attack effectiveness.
- Cross-domain analysis: We compare adversarial robustness between raw I/Q-based and TFI-based detectors in both white-box and black-box settings.
- Modulation-wise robustness: We conduct a detailed per-modulation analysis across twelve LPI waveform types, revealing distinct vulnerability patterns and robustness characteristics.

II. I/Q-Based LPI Radar Detection

2.1 LPI Signals

LPI radars are designed to evade detection and analysis by an intercept receiver. Modern systems pursue this goal by varying carrier settings and modulation formats while operating at comparatively low power, which complicates interception. Typical LPI emissions combine low peak pulse power, broad spectral occupancy, narrow beams, and subtle intra-pulse structure. Under these conditions, straightforward FFT-centric detectors and classifiers often lose effectiveness [1].

In this work, the received complex baseband is modeled as,

$$x(t) = Ae^{j(2\pi f_c t + \phi(t))} + n(t) \quad (1)$$

where A is the amplitude, f_c is the carrier

frequency, $\phi(t)$ is the intra-pulse phase trajectory, and $n(t)$ is complex additive white Gaussian noise (AWGN). This representation mirrors what an intercept receiver processes and separates the deterministic waveform component from the stochastic disturbance. Discrimination among LPI waveforms is largely determined by ambiguity-function and autocorrelation behavior, which are governed by the choice of intra-pulse modulation [15]. For taxonomy, we adopt the conventional division into frequency-modulated (FM) and phase-modulated (PM) families. FM signals adjust their instantaneous frequency over the pulse, for example linear FM chirps, Costas frequency hopping, and nonlinear FM. PM signals shape the phase sequence while holding the carrier fixed, for example biphasic and polyphase codes. Viewed equivalently, FM varies the frequency term under a fixed phase law, whereas PM varies the phase term with a constant carrier.

Following common recognition benchmarks, our corpus includes twelve representative intra-pulse modulations: two FM types (LFM and Costas) and ten PM types (BPSK, Frank, P1, P2, P3, P4, T1, T2, T3, T4) [9]. These classes span widely used frequency-hopping, biphasic, and polyphase designs in LPI radar and provide a standard basis for evaluating detection and classification performance.

2.2 LPI detection model

This study assumes a realistic time-domain LPI intercept receiver and uses it as a reference to analyze vulnerability to adversarial perturbations constrained by PSR and PNR [10]. Accordingly, this section introduces the baseline detection model that serves as the reference interceptor throughout the adversarial analysis.

In an LPI radar scenario, an intercept

receiver observes incoming signals and attempts to determine the presence of a radar emission. Beyond mere detection, a critical function of the interceptor is to identify the intra-pulse modulation scheme, which provides essential information for radar classification, threat assessment, and subsequent localization or tracking processes. In this work, this task is formulated as a multi-class modulation classification problem, and the corresponding classifier is treated as the detector.

To define adversarial attacks in a principled manner, the input-output relationship and decision boundaries of the detector must be explicitly specified. We therefore assume a neural-network-based detector that directly ingests time-domain I/Q signals and outputs modulation labels. This assumed detector forms the basis for defining the attacker-detector interaction in both white-box (WB) and black-box (BB) settings [12].

We adopt a time-domain detection framework that operates directly on raw complex baseband I/Q signals, without converting them into time-frequency representations such as STFT-based images. This choice is not merely an implementation preference, but is closely tied to the goals of the adversarial analysis [16]. Distortion constraints such as PSR and PNR are physically meaningful metrics defined in terms of signal and noise energy. These quantities are most naturally enforced and interpreted in the original signal domain.

Specifically, we consider four time-domain models: RawNet [7], ResNet1D, a GRU-based model [17], and a Transformer-based model [18], which respectively capture local convolutional features, deep residual representations, temporal dependencies, and long-range interactions. In addition, for comparison with conventional approaches, we

include TFI-based detectors using ResNet50 [19] and Vision Transformer (ViT) [20], which operate on spectrogram-like representations.

The input to the detector is a received complex baseband signal represented by its in-phase (I) and quadrature (Q) components. Each input sample is expressed as $x \in \mathbb{R}^{2 \times N}$, where N denotes the signal length. The detector maps this input to one of twelve LPI modulation classes, corresponding to the modulation scheme inferred by the intercept receiver. The detection model defined in this section is used as the attack target in Section 3. In the adversarial setting, it represents the detector whose decision boundaries are either fully accessible (WB) or approximated via a surrogate model (BB), thereby providing a consistent reference for evaluating adversarial effectiveness under PSR/PNR constraints.

III. Adversarial Attacks

3.1 Threat Model and Attack Formulation

While LPI signal detection is critical for intercept receivers, from the perspective of an LPI radar operator, it is equally important to avoid detection and classification by adversarial receivers. In practical electronic warfare scenarios, an LPI radar system must not only remain difficult to detect, but also resist accurate modulation classification by an interceptor.

In this work, we investigate adversarial perturbations that minimally modify the transmitted signal while effectively degrading the performance of interception systems. Specifically, we focus on signal-domain perturbations applied to raw complex-valued I/Q sequences.

Let x denote the original signal and δ the adversarial perturbation. The adversarial signal

is defined as $x_{adv} = x + \delta$, and the objective is to degrade the detector's classification performance.

We consider untargeted attacks, where the goal is to induce misclassification rather than enforce a specific incorrect label. This setting is particularly appropriate for LPI scenarios, as any incorrect classification can disrupt subsequent processes such as radar identification, threat assessment, and emitter localization.

To analyze adversarial behavior, we consider both WB and BB threat models. In the WB setting, the attacker has full access to the detector, including model parameters and gradients. In the BB setting, the attacker relies on a surrogate model to generate transferable perturbations, reflecting realistic conditions where the exact interceptor model is unknown [21].

3.2 Attack Methods and Constraint Modeling

We consider four gradient-based adversarial attack methods: FGSM, PGD, MI-FGSM, and MI-DI-FGSM. FGSM generates an adversarial

constraints, step size, detector and loss function respectively. PGD extends this approach by iteratively updating the perturbation with projection onto a feasible constraint set:

$$x^{(k+1)} = \text{Proj}_C(x^{(k)} + \eta \text{sign}(\nabla_{x^{(k)}} L(f(x^{(k)}), y)))$$

$$, k = 0, \dots, K-1,$$

with initialization $x^{(0)} = x$. Here, η denotes the step size and the projection ensures feasibility with respect to constraint C at every iteration.

MI-FGSM introduces a momentum term to stabilize gradient updates and improve convergence. MI-DI-FGSM further enhances attack robustness by incorporating input diversity. While input diversity was originally proposed for image-based attacks, we extend it to the raw signal domain by designing signal-specific transformations. These include random time shifting, random cropping with repetition padding, light noise injection, and phase rotation, which introduce controlled variability while preserving essential signal characteristics.

Unlike image-based adversarial attacks that constrain perturbations using pixel-level bounds

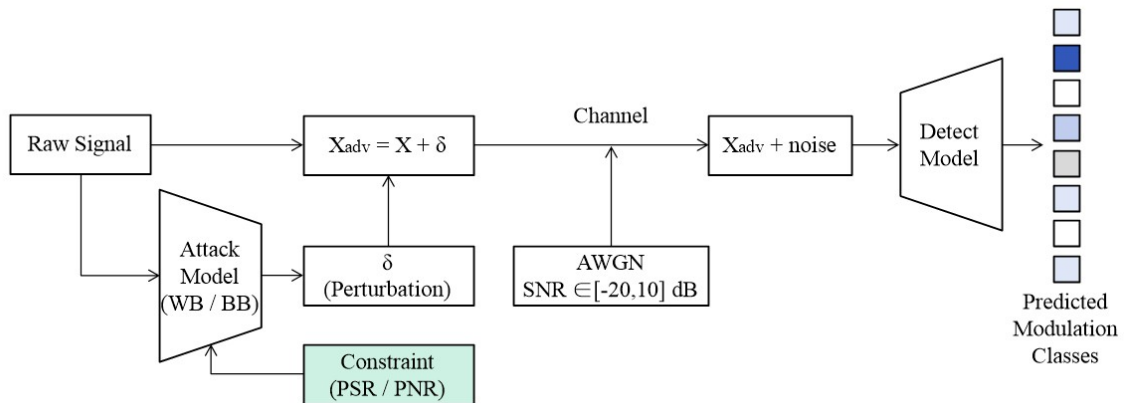


Fig. 1. Adversarial attack pipeline for raw I/Q LPI signal classification, including perturbation generation under PSR/PNR constraints, channel modeling, and detector inference.

perturbation by performing a single-step update along the gradient direction:

$$x_{adv} = \text{Proj}_C(x + \alpha \text{sign}(\nabla_x L(f(x), y))),$$

where C , α , $f(\cdot)$, and $L(\cdot)$ denotes

(e.g., ϵ), signal-domain attacks require physically meaningful distortion measures. In this work, we adopt PSR and PNR as constraints, following prior work in radio signal

classification [10].

PSR measures the relative energy of the perturbation with respect to the original signal, while PNR quantifies the perturbation strength relative to noise. The PSR is defined as

$$\text{PSR}[\text{dB}] = 10\log_{10} \frac{\|\delta\|_2^2}{\|x\|_2^2},$$

and the PNR is defined as

$$\text{PNR}[\text{dB}] = 10\log_{10} \frac{\|\delta\|_2^2}{\|n\|_2^2},$$

where x is the clean signal, δ is the adversarial perturbation, and n denotes channel noise. These quantities satisfy $\text{PNR} = \text{PSR} + \text{SNR}$. Accordingly, the feasible perturbation set is defined as

$$C = \{\delta \mid \text{PSR} \geq \tau_{\text{PSR}}, \text{PNR} \geq \tau_{\text{PNR}}\},$$

where τ_{PSR} and τ_{PNR} denote predefined threshold values that control the minimum allowable perturbation analysis.

These constraints allow controlled manipulation of signal distortion. A large distortion budget can significantly degrade classification performance but may distort the signal excessively, while overly small perturbations may fail to induce meaningful misclassification. Therefore, PSR and PNR provide a principled balance between attack effectiveness and signal fidelity.

Table 1. Summary of Detector Architectures and Model Configurations

Model	Input Shape	Main Layers	
RawNet [7]	2×1600	Conv1D×13, BN×13, FC×1	Re
ResNet1D [19]		Conv1D×20, BN×20, FC×1	Bas
GRU [17]		Conv tokenizer×3, FC×4	2-l atte
Transformer [18]		Conv tokenizer×3, FC×10	Transf MultiH
ResNet50 [19]	3×224×224	Conv2D, BN, FC	Bottl
ViT [20]		Linear, MLP	Transf MultiH

3.3 Adversarial Evaluation Pipeline

Fig. 1 illustrates the overall adversarial attack pipeline in the raw I/Q domain. Given a clean input signal x , an adversarial perturbation δ is generated using gradient-based methods under PSR/PNR

constraints. The adversarial signal $x_{adv} = x + \delta$, is then passed through a channel model with additive noise, and the resulting signal is fed into the detector for classification.

This pipeline reflects a realistic interception scenario in which perturbations are applied at the transmitter and evaluated after channel propagation. It enables a consistent and physically meaningful assessment of adversarial effectiveness under practical signal conditions.

By optimizing perturbations under such variability, the resulting adversarial signals become more robust to channel effects and more effective when transferred to unseen models, enabling realistic evaluation of adversarial attacks in LPI scenarios.

IV. Experimental Setup

4.1 Dataset Configuration

The experimental corpus consists of twelve representative LPI intra-pulse modulation types, including two frequency-modulated (LFM, Costas) and ten phase-modulated (BPSK, Frank,

P1-P4, T1-T4) waveforms. Each sample is represented as a raw complex-valued baseband signal composed of I/Q components. The input to the detector is therefore a two-channel real-valued tensor $x \in \mathbb{R}^{2 \times N}$, where N denotes the signal length. For detector pre-training, a large-scale dataset comprising approximately 1.2 million samples was generated.

The data are evenly distributed across modulation classes and span an signal-to-noise ratio (SNR) range from -20 dB to 10 dB in 2 dB increments. To ensure realistic variability, waveform parameters such as carrier frequency, phase coding, and pulse duration are randomly sampled within predefined ranges for each modulation scheme. The dataset is split into training, validation, and test sets with a ratio of 8:1:1. To avoid training-test leakage, adversarial evaluation is conducted on a separate held-out dataset consisting of 120,000 samples (10,000 per modulation class), with SNR ranging from -10 dB to 10 dB. All adversarial experiments are performed exclusively on this evaluation set.

4.2 Detector Architectures and Training Details

To evaluate model-dependent adversarial vulnerability, we consider multiple time-domain I/Q-based detectors with diverse architectural paradigms, including convolutional, recurrent, and attention-based models, as described in Section 2.2. Table 1 summarizes the architectural configurations and key parameters of all evaluated detectors.

Although the models differ significantly in their internal structures, they share a unified input-output interface. Specifically, all models operate directly on raw complex-valued I/Q sequences without time-frequency transformation and predict modulation classes

over an identical label space. This unified setting ensures a fair comparison, allowing performance differences to be attributed solely to architectural characteristics rather than input representation or task definition.

All detectors are trained using the Adam optimizer with cross-entropy loss. The initial learning rate is set to 1×10^{-3} and decayed to 1×10^{-5} using a cosine annealing schedule. The batch size is 64, and training is conducted for 100 epochs. Model selection is performed based on validation accuracy.

Input signals are normalized via global amplitude scaling to stabilize optimization while preserving relative energy relationships required for PSR/PNR computation. This normalization ensures that distortion constraints remain physically meaningful during adversarial attack generation.

4.3 Adversarial Attack Configuration

We evaluate adversarial attacks under both white-box and black-box threat models, with explicit implementation strategies designed to reflect realistic interception scenarios.

In the WB setting, the attacker has full access to the detector, including model architecture, parameters, and gradients. Accordingly, the same trained model is used for both detection and adversarial perturbation generation, ensuring that the attack is optimized directly against the target detector.

In the BB setting, the attacker does not have access to the defender's parameters. To simulate this scenario, we employ two configurations. First, for same-model BB evaluation, the attacker and defender share the same network architecture but are trained independently using different random seeds and separately generated datasets. This results in

different learned weights while preserving identical model structure and label space. Second, to evaluate cross-model transferability, we perform cross-architecture BB attacks, where perturbations are generated on one model and evaluated on a different model with a distinct architecture. These configurations allow us to systematically analyze both intra-architecture and inter-architecture transferability.

To further improve transferability under realistic channel conditions, EOT [14], [22] is applied in the BB setting. At each gradient step, the loss is averaged over multiple stochastic transformations sampled from a predefined transformation set. Specifically, we consider AWGN and Rayleigh fading channels, carrier frequency offsets within $\pm 1\%$, timing offsets within ± 2 samples, and minor amplitude scaling perturbations. In practice, 8 transformation samples are used per gradient step.

These configurations ensure that adversarial perturbations are evaluated not only under idealized WB conditions but also under practical BB scenarios with channel variability and model uncertainty.

4.4 Attack Methods and Hyperparameters

All adversarial attacks are conducted in an untargeted setting under explicit PSR or PNR constraints. The distortion budget is varied from -20 dB to 0 dB in 2 dB increments. During evaluation, adversarial performance is measured across an SNR range from -10 dB to 10 dB in 2 dB increments.

We consider four gradient-based attack methods: FGSM, PGD, MI-FGSM, and MI-DI-FGSM. For FGSM, a single gradient step is applied with step size α , followed by projection to satisfy the target PSR or PNR constraint. Due to its single-step nature, FGSM

serves as a baseline for evaluating the effectiveness of iterative attacks.

For PGD, iterative updates are performed with projection onto the feasible perturbation set at each step. The number of iterations is set to $K=20$, and the step size is scaled proportionally to the distortion budget. Unless otherwise specified, the attack is initialized from the clean signal without random initialization.

For MI-FGSM, a momentum term is introduced to stabilize gradient updates and improve attack effectiveness. The decay factor for momentum is set to 1.0 , and gradients are accumulated across iterations with momentum before applying the update.

For MI-DI-FGSM, input diversity is incorporated to improve generalization and transferability. At each iteration, stochastic signal-domain transformations are applied with a probability of 0.7 . The transformations include (1) random time shifting within ± 10 samples, (2) random cropping of $80\text{--}100\%$ of the signal length followed by repetition padding to restore the original length, (3) additive Gaussian noise with a standard deviation of 0.01 relative to the signal amplitude, and (4) phase rotation with an angle uniformly sampled from $[-\pi/12, \pi/12]$. These transformations introduce controlled variability into the input while preserving key signal structures relevant to LPI detection, resulting in more robust and transferable adversarial perturbations.

All iterative attacks enforce projection onto the feasible PSR/PNR constraint set at every iteration, ensuring that perturbations strictly satisfy the specified distortion constraints throughout the attack process.

4.5 Evaluation Metrics

The primary evaluation metric in this study is the Attack Success Rate (ASR), which measures

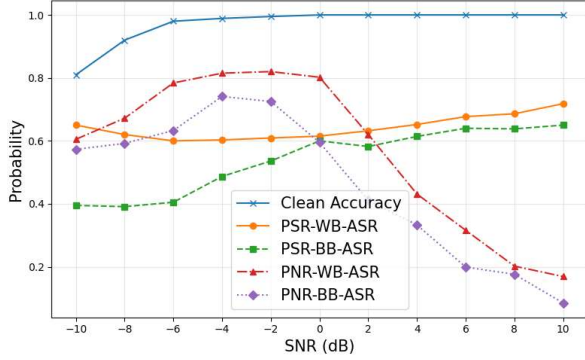


Fig. 2. Probability vs. SNR with clean accuracy and ASR under fixed distortion budgets (PSR = -10 dB, PNR = -10 dB).

the proportion of originally correctly classified samples that become misclassified after adversarial perturbation [23]. Let $\{(x_i, y_i)\}_{i=1}^M$ denote the attack dataset and $f(\cdot)$ the trained detector. We first define the clean-correct subset as $S = \{i \mid f(x_i) = y_i\}$. The ASR is computed as

$$ASR = \frac{\sum_{i \in S} 1[f(x_i^{adv}) \neq y_i]}{|S|},$$

where x_i^{adv} denotes the adversarial example and $1[\cdot]$ is the indicator function. This definition ensures that attack effectiveness is evaluated only on samples that are correctly classified under clean conditions, thereby isolating the degradation caused by adversarial perturbations [23].

V. Experimental Results and Analysis

In this section, we present a comprehensive evaluation of adversarial attacks on raw I/Q-based LPI radar signal classification. We analyze the effects of SNR, perturbation constraints, attack methods, optimization strategies, domain representation, and model transferability.

Figure 2 illustrates how SNR affects the probability of the outcome curves, plotting

clean accuracy together with the ASR probabilities using PGD-based attacks under fixed distortion budgets of PSR = -10 dB and PNR = -10 dB, for both WB and BB settings. Under the PSR constraint,

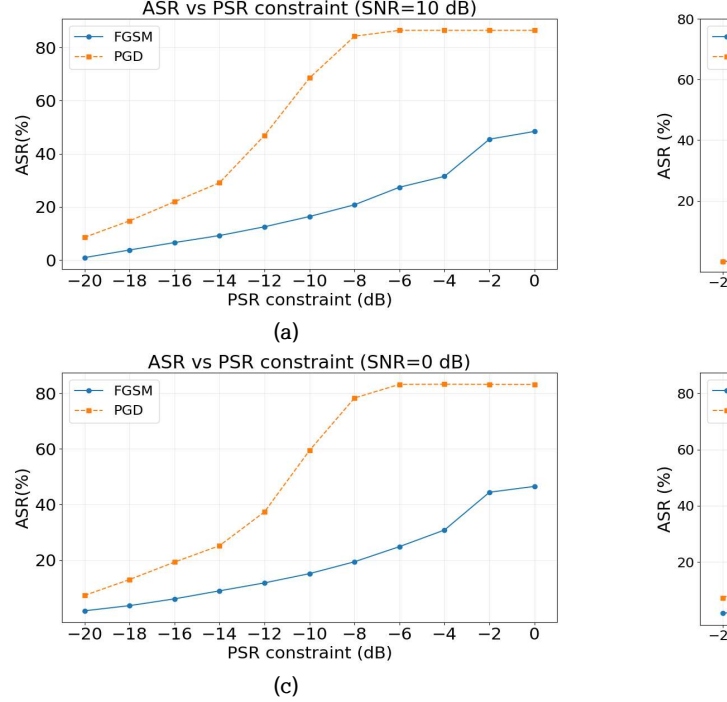


Fig. 3. Effect of PSR/PNR Constraints on ASR under Two SNR Regimes (SNR=10 dB and SNR=0 dB).

the ASR probability increases with SNR for both WB and BB, reaching a peak or approaching saturation in the mid-to-high SNR regime. This behavior is consistent with $PNR=PSR+SNR$, which makes the perturbation stronger relative to noise as SNR grows. In contrast, under the PNR constraint, the ASR probability decreases monotonically with SNR, most sharply for BB, since $PSR=PNR-SNR$ implies a rapidly diminishing perturbation relative to the signal.

Across all conditions, WB outperforms BB, reflecting the inherent advantage of gradient access. The gap between WB and BB widens at high SNR in the PNR-constrained setting where transferability deteriorates. Taken together, these trends indicate that PSR-constrained attacks are most consequential at moderate to high SNR, whereas PNR-constrained attacks are

comparatively potent at low SNR. These findings motivate defenses that combine preprocessing and channel randomization to further erode BB transfer in high-SNR, PNR-constrained regimes, together with stronger robustness measures such as adversarial training against PSR-constrained threats.

Figure 3 characterizes how ASR varies with the distortion budget under two constraints, namely PSR and PNR, at SNR = 10 dB (Fig. 3 (a) and (b)) and SNR = 0 dB (Fig. 3 (c) and (d)). In each figure we compare the FGSM (solid line) and PGD (dotted line). Across all settings, ASR increases monotonically as the budget relaxes from -20 dB to 0 dB, and PGD consistently exceeds FGSM, reflecting the advantage of multistep optimization. Under the PSR constraint (Fig. 3 (a) and (c)), higher SNR yields higher ASR because $\text{PNR} = \text{PSR} + \text{SNR}$ makes the perturbation more prominent relative to noise; at SNR = 10 dB, PGD shows a pronounced knee near PSR around -10 dB and then approaches saturation above approximately -8 dB, while FGSM rises more gradually. On the other hand, under the PNR constraint (Fig. 3 (b) and (d)), the opposite trend appears: higher SNR suppresses ASR since $\text{PSR} = \text{PNR} - \text{SNR}$ forces the perturbation to become smaller relative to the signal; in here, PGD again dominates, showing a sharp transition as the PNR

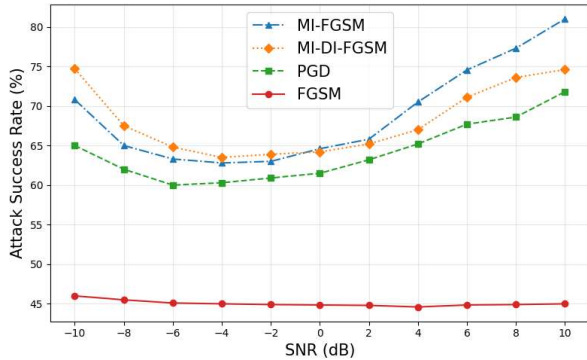


Fig. 4. ASR across SNRs for different attack methods (FGSM, PGD, MI-FGSM, and MI-DI-FGSM) under a fixed PSR constraint of -10 dB in the WB setting.

budget crosses -10 dB, whereas FGSM increases steadily but remains well below PGD. Overall, the figure 3 shows that PGD is uniformly stronger than FGSM, PSR-constrained attacks benefit from higher SNR, and PNR-constrained attacks are comparatively more effective at lower SNR.

Fig. 4 compares the effectiveness of different attack methods under a fixed PSR constraint of -10 dB. By enforcing the same perturbation budget, this comparison evaluates how effectively each method exploits the allowable distortion to degrade detection performance. FGSM yields negligible ASR across all SNR levels, confirming that single-step perturbations are insufficient in the raw I/Q domain. In contrast, PGD significantly improves attack performance through iterative refinement. Incorporating momentum further enhances effectiveness, as demonstrated by MI-FGSM, while MI-DI-FGSM introduces signal-domain diversity transformations such as time shifts, cropping with repetition padding, light noise injection, and phase rotation. Although this diversity slightly reduces white-box effectiveness, it provides a more generalized formulation and is expected to improve transferability. Overall, these results highlight that attack effectiveness is strongly dependent on the optimization strategy under a fixed perturbation budget.

Fig. 5 presents the classification accuracy and ASR for different modulation types. The results

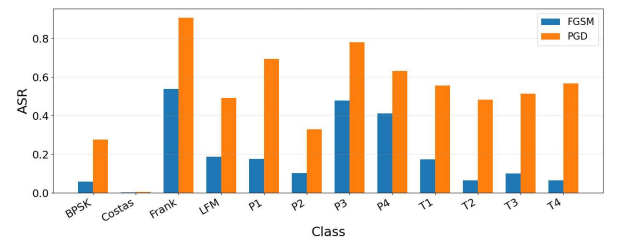


Fig. 5. Modulation-wise ASR at SNR = 0 dB under PSR-constrained black-box attacks.

reveal clear modulation-dependent vulnerability.

In particular, polyphase and T-code families tend to exhibit higher ASR, whereas Costas and BPSK remain comparatively robust. This suggests that intrinsic waveform structure plays a key role in adversarial sensitivity.

Table 2 presents a 12×12 confusion matrix over the LPI modulation set at SNR = 0 dB in a black box PGD setting, where the PSR budget is averaged over the range -20 to 0 dB. Rows denote true classes and columns denote predicted classes. Each cell reports two values: one is for the clean proportion and the other is for the post attack proportion (in parentheses) on the same test split. The main diagonal therefore captures the per class probability of correct detection, and off diagonal entries represent misclassification patterns. The attack reduces most diagonal entries, confirming a class dependent degradation in detection probability, while simultaneously

Table 2. Class-wise confusion matrix at SNR = 0 dB under PSR-constrained black-box PGD attacks, averaged over PSR $\in [-20, 0]$ dB.

	BPSK	Costas	Frank	LFM	P1	P2	P3	P4	T1	T2	T3	T4
BPSK	1.00 (0.72)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.11)	0.00 (0.16)	0.00 (0.01)	0.00 (0.00)
Costas	0.00 (0.00)	1.00 (1.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Frank	0.00 (0.04)	0.00 (0.00)	0.99 (0.09)	0.00 (0.00)	0.00 (0.04)	0.00 (0.00)	0.01 (0.82)	0.00 (0.01)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
LFM	0.00 (0.00)	0.00 (0.00)	0.00 (0.04)	1.00 (0.51)	0.00 (0.00)	0.00 (0.00)	0.00 (0.06)	0.00 (0.40)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
P1	0.00 (0.00)	0.01 (0.00)	0.00 (0.15)	0.00 (0.00)	0.99 (0.31)	0.00 (0.00)	0.00 (0.00)	0.00 (0.50)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.04)
P2	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.99 (0.67)	0.00 (0.00)	0.00 (0.00)	0.01 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.33)
P3	0.00 (0.00)	0.00 (0.00)	0.00 (0.55)	0.00 (0.01)	0.00 (0.00)	0.00 (0.00)	1.00 (0.22)	0.00 (0.21)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
P4	0.00 (0.00)	0.00 (0.00)	0.00 (0.09)	0.00 (0.00)	0.00 (0.32)	0.00 (0.00)	0.00 (0.22)	1.00 (0.37)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
T1	0.08 (0.07)	0.00 (0.00)	0.07 (0.05)	0.00 (0.00)	0.01 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.99 (0.44)	0.00 (0.03)	0.00 (0.03)	0.00 (0.38)
T2	0.00 (0.36)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.01)	1.00 (0.52)	0.00 (0.00)	0.00 (0.11)
T3	0.00 (0.01)	0.00 (0.00)	0.00 (0.01)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.01 (0.00)	0.00 (0.46)	0.00 (0.00)	0.99 (0.49)	0.00 (0.02)
T4	0.00 (0.05)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.12)	0.00 (0.02)	0.00 (0.00)	0.00 (0.02)	0.00 (0.30)	0.01 (0.03)	0.00 (0.03)	0.99 (0.43)

Table 3. Effect of PGD Iterations on Attack Success Rate under PSR = -10 dB

Steps	ASR(%)
1	12.15
5	26.69
10	56.44
20	75.32
40	79.26

several family proximal confusions. Two trends are prominent. First, Costas and BPSK remain comparatively robust: their diagonal values drop the least and their off diagonal growth is limited, which is consistent with the class wise ASR analysis. Second, the largest misclassification increases occur within families. Polyphase codes exhibit strong mutual confusions such as P1 to P2 and P3 to P4, and the T code set shows similar behavior between adjacent members such as T1 to T2 and T3 to T4. LFM also leaks into early polyphase classes, indicating that the perturbation tends to imprint phase like signatures that shift decisions toward nearby phase structure patterns rather than across families. Several confusions are asymmetric, for example P3 to P4 being more frequent than the reverse, which suggests decision boundaries that are easier to traverse from one side.

To further analyze the role of iterative optimization, we vary the number of PGD iterations

while fixing the PSR constraint at -10 dB. The results are summarized in Table 3. As shown in Table 3, when the number of iterations is limited to a single step, the attack reduces to a one-step method similar to FGSM, resulting in a low ASR of 12.15%. As the number of iterations increases, the ASR improves significantly, reaching 26.69% at 5 steps, 56.44% at 10 steps, 75.32% at 20 steps, and 79.26% at 40 steps.

Since all experiments are conducted under the same PSR constraint, the performance gain originates purely from improved optimization rather than increased perturbation energy. Notably, the

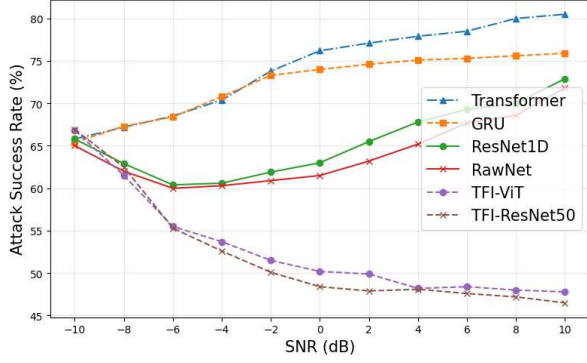


Fig. 6. ASR comparison between raw I/Q-based and TFI-based detectors across SNR levels under PSR-constrained PGD attacks.

improvement saturates as the number of iterations increases, indicating diminishing returns and highlighting a practical trade-off between attack strength and computational cost.

Fig. 6 highlights a fundamental difference between raw I/Q-based and TFI-based representations. Before discussing the results, it is important to clarify that the two domains are evaluated under different perturbation constraints. For raw I/Q-based models, adversarial attacks are conducted under a fixed PSR constraint of -10 dB, which is a physically meaningful measure in the signal domain. However, this constraint cannot be directly applied to TFI-based models, as they operate in the image domain. Therefore, for TFI-based models, we adopt an ϵ -bounded attack formulation. Following our previous study [9], we set $\epsilon = 8/255$, which provides a sufficiently strong perturbation while preserving the structural characteristics of the TFI representation.

Overall, TFI-based models exhibit lower ASR compared to raw I/Q-based models. This can be attributed to the inherent robustness of image-based feature extraction, where local spatial aggregation and normalization tend to smooth out small perturbations. In contrast, raw I/Q-based models operate directly on fine-grained signal variations, making them

more sensitive to adversarial perturbations under the same relative constraint.

Within the raw signal domain, CNN-based models such as RawNet and ResNet1D show similar performance, suggesting that conventional convolutional and residual structures have limited capability in capturing long-range dependencies that may be exploited by adversarial perturbations. In contrast, sequence modeling approaches such as GRU and Transformer exhibit consistently higher ASR, indicating greater vulnerability.

In particular, the Transformer-based model shows the highest ASR across most SNR levels. This behavior can be attributed to its global self-attention mechanism, which aggregates information across the entire signal sequence. While this enables powerful representation learning, it also allows adversarial perturbations to propagate more effectively across the input, leading to stronger attack impact.

Taken together, the results presented in Figs. 2–6 and Tables 2–3 lead to four main observations. First, adversarial effectiveness is strongly influenced by both SNR and perturbation constraints, with PSR-constrained attacks becoming more effective at higher SNR levels. Second, iterative optimization plays a critical role, as PGD and its variants significantly outperform single-step FGSM. Third, adversarial vulnerability is highly dependent on the signal representation domain, with raw I/Q and TFI models exhibiting distinct SNR-dependent behaviors.

VI. Conclusion

This paper presented a systematic adversarial evaluation framework for time-domain I/Q-based LPI signal detection under physically meaningful PSR and PNR constraints. Through

extensive experiments across SNR regimes, attack methods, and modulation types, we demonstrated that adversarial perturbations can significantly degrade detection performance even under strict signal-domain distortion limits.

Among the evaluated methods, iterative attacks consistently outperformed single-step approaches, with MI-based variants further improving attack effectiveness and transferability. In particular, the incorporation of input diversity in the raw signal domain was shown to enhance robustness against channel variability and model mismatch. The impact of constraint type was also clearly observed: PSR-constrained attacks became increasingly effective at higher SNR, whereas PNR-constrained attacks were more dominant in low-SNR regimes.

In addition, we revealed that adversarial vulnerability is highly dependent on modulation structure. Polyphase and T-code waveforms exhibited greater susceptibility, while Costas and BPSK remained relatively robust. Misclassification patterns were predominantly observed within modulation families, indicating that adversarial perturbations tend to exploit intrinsic structural similarities.

These findings provide practical insights for the design of robust intercept receivers. First, defense strategies should be context-aware: lightweight randomization and preprocessing diversity can mitigate weak transfer attacks, while stronger regimes require adversarial training and margin-based defenses. Second, signal-domain constraint modeling using PSR and PNR offers a principled and interpretable framework for balancing attack effectiveness and signal fidelity.

Overall, this work establishes a comprehensive baseline for adversarial analysis

in I/Q-based LPI detection and highlights critical differences between raw signal and TFI-based representations. Future work will extend this framework to unified comparisons across representation domains and explore defense mechanisms under realistic over-the-air conditions.

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